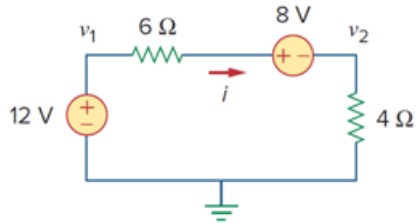


Problem

For the circuit in Fig. 3.47, v_1 and v_2 are related as:

- (a) $v_1 = 6i + 8 + v_2$
- (b) $v_1 = 6i - 8 + v_2$
- (c) $v_1 = -6i + 8 + v_2$
- (d) $v_1 = -6i - 8 + v_2$

Figure 3.47



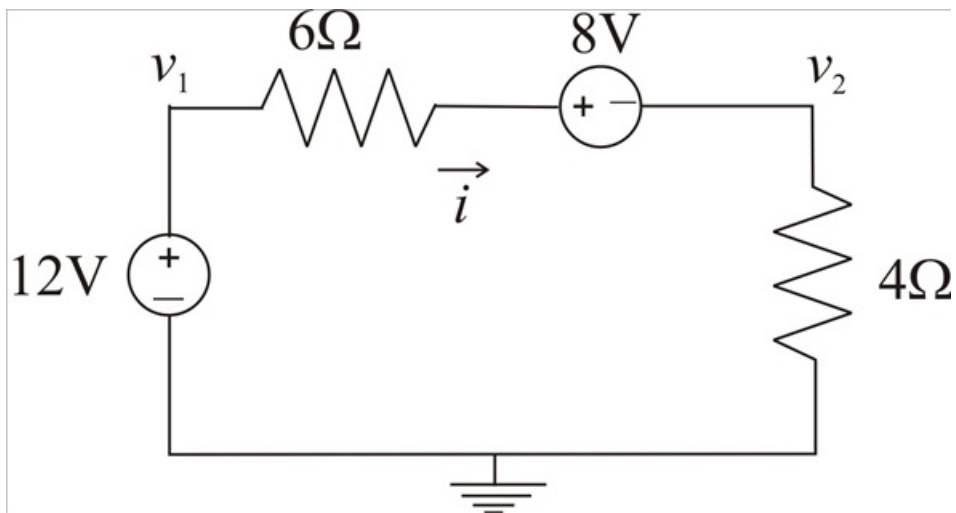
Step-by-step solution

Step 1 of 3

Answer: (a)

Step 2 of 3

Given circuit diagram



Step 3 of 3

By applying KVL to the above circuit we can get

$$-v_1 + 6i + 8 + v_2 = 0$$

$$\boxed{v_1 = 6i + 8 + v_2}$$